Concise 300 Circular Slide Rule

someonesdad1@gmail.com 27 Aug 2012, updated 2 Aug 2026

This document looks at the Concise 300 circular slide rule, one of the few slide rules that is still in production. As of August 2026, they are still being made in Japan.

Updates

3 Apr 2023: Minor updates, remove broken links.

2 Aug 2026: Removed examples to shorten to 5 pages

What is it?

The Concise 300 is a circular slide rule 110 mm in diameter, roughly the width of your hand. It has a mass of 48 g. The body of the rule is 3.6 mm thick and the maximum thickness of the rule is over the rivet, which is 9 mm at the thickest part. The thickness over the transparent cursor is 5.2 mm.

Here's a picture of the rule (picture from [*srm*]):

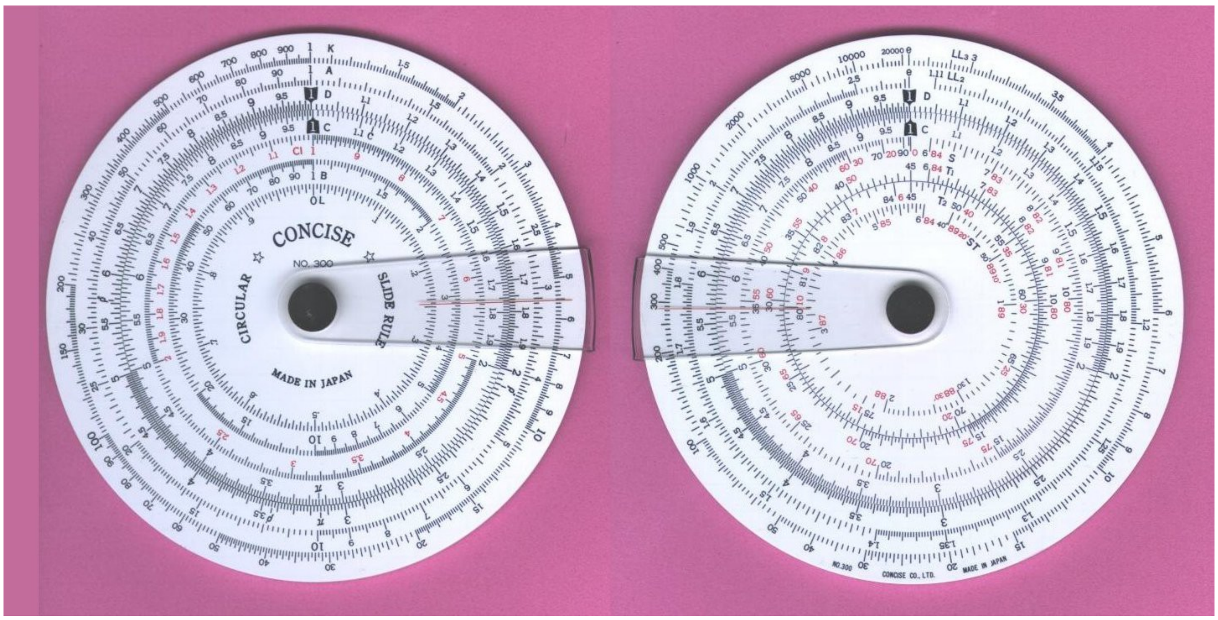


Figure 1

The diameter of the C and D scales is 80 mm, meaning their linear length is 80π mm or 251 mm or 9.9 inches. This means the resolution of the slide rule is the same as the typical 10 inch linear slide rule. But it's more compact than a 10 inch linear slide rule and can slip into a shirt pocket.

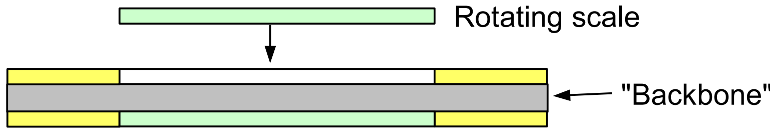
Each side of the rule consists of a fixed portion; the two fixed portions contain the D, A, and K scales on the front and D, LL2, and LL3 scales on the back. Each side has a rotating disk; both rotating disks have a C scale at their OD that meshes with the D scale to provide for multiplication and division. It requires good manufacturing to get a close and concentric fit between the rotating scale and the fixed scale.

The two sides of the rule are independent and not typically used together. If you put the cursor on the 1 of the D scale on one side, it will be lined up with the 1 on the D scale of the other side. Because the D scale has its numbers increase clockwise on each side, you can theoretically set a number on one D scale, then flip the rule over and read the reciprocal on the other scale. I say theoretically because in practice, this doesn't work as well as using the C and CI scales for reciprocals. For example, on my 300, setting the cursor over 2.5 and flipping the rule gives 3.99 on the other scale (it should be 4.00).

Construction

The rule appears to be made from a sheet plastic like PVC. The construction is good and, with care, these slide rules should work indefinitely as long as the plastic doesn't degrade or an inappropriate solvent isn't spilled on the device. The graduations and lettering appear to be chemically etched and the indentations are filled with black or red coloring. Thus, moving your fingers over the rule should never wear the markings off like can happen with cheaper rules that just print the markings on the surface.

If I had to guess how they're made, there are two thicknesses of PVC sheet: 0.8 mm and 1.5 mm. The main body would be constructed by gluing two pieces shaped in an annular form and a disk together; the side view would be



The yellow annulus is 0.8 mm thick and provide the relief for the inner rotating scales (light green) on each side. The gray disk is 1.5 mm thick and provides the slide rule's rigidity. The two yellow annuli are glued to the backbone.

The two light green rotating disks are 80 mm in diameter and are 0.8 mm thick.

The slide rule is held together by a metal assembly (I'll call it a rivet) that is probably pressed together and permanently assembles itself through some metal deformation. Each rotating disk has a wavy washer to provide friction along with what appears to be a thin (perhaps 0.1 mm thick) PVC washer to insulate the rotating disk from the wavy washer. The metal rivet also holds the clear plastic cursor, which is a piece that is bent over 180° and has hairlines scribed and filled on them (with red paint or ink). I suspect the primary purpose of the wavy washers is to provide suitable friction for the cursor.

The plastic cursor is made from a transparent plastic about 0.8 mm thick. The edges have been milled and the cursor was probably bent while warm into the final position. It's likely either polycarbonate or acrylic. Because of the way a small section of it scratched with a small screwdriver, I would guess it's acrylic.

Once this slide rule is assembled correctly, there's basically nothing that can go wrong with it after that (excluding a design flaw). Concise has been making slide rules since 1949, so I'll assume they've got the process down pat. The only things that I can envision that would damage these slide rules would be:

1. Excessive heat. The manual says to not expose them to temperatures greater than 60 °C (140 °F).
2. Some organic solvents.
3. Strong acids or bases or acidic/basic vapors.
4. Being left out in the sun which can cause UV discoloration and embrittlement if the plastic isn't UV-stabilized.
5. Mechanical damage (being run over by a car, chewed on by a dog, abrasion, etc.).
6. Strong nuclear radiations.
7. Being put into a microwave oven.

I would imagine that the rule could be resistant to having things like coffee or a soft drink spilled on it. It would probably clean up well in hot (no more than 60 °C), clean tap water by rinsing a number of times (a mild detergent wouldn't hurt anything, but I'd use the plain water first to see if it did the job).

Inspection of the marks and lettering with 4X and 10X loupes show that the marks are very clean and well-made. I would assume that the method involves an etching process, as etching technology has been well-developed for the semiconductor industry. At 10X magnification, the lettering appears to have been made by a computer image.

The finish of the rule's surface is predominantly matte, but there's a small amount of specular reflection. It's not annoying enough to interfere with the use of the rule.

Slide rule scales

Here I describe the scales on each side of the rule, starting at the outermost scale and working in towards the center.

Side 1

Outermost scales on fixed part of rule and their use (x is the setting on the C or D scale):

K	x^3
A	x^2
D	General multiplication, division, ratios

Innermost scales on rotating part of rule and their use (x is the setting on the C or D scale):

C	General multiplication, division, ratios
CI	Reciprocals
B	x^2
L	log(x)

Side 2

Outermost scales on fixed part of rule and their use (x is the setting on the C or D scale):

LL3	e^x
LL2	$e^{0.1x}$
D	General multiplication, division, ratios

Innermost scales on rotating part of rule and their use (x is the setting on the C or D scale):

C	General multiplication, division, ratios
S	$\sin^{-1}(x)$, for x from 0.1 to 1
T1	$\tan^{-1}(x)$, for x from 0.1 to 1
T2	$\tan^{-1}(x)$, for x from 1 to 10
ST	$\sin^{-1}(x)$ and $\tan^{-1}(x)$, for x from 0.01 to 0.1

Intended users

This is a slide rule that is aimed mostly at technical workers. The reason is because of the addition of the LL2 and LL3 scales; these scales are used to raise numbers to a power and get natural logarithms. This is a common task for engineers and scientists.

The other feature that is used by technical workers is the ability to calculate trigonometric functions. These are commonly used in the solution of triangles. If you don't need the LL scales but need the trig functions, the Concise model 270 may be a better choice, as it has some extra scales for trigonometry calculations.

Opinions

Likes

The construction of this rule is first rate. I cannot see any reasonable way of improving the existing mechanical design. Unless the cursor yellows badly over time, I would assume the rule will last indefinitely. The secret would be to store the rule out of the light; its cheap vinyl case is suitable for this.

An engineering refinement would be to design a central rivet that could be squeezed with the two fingers. When squeezed, the rivet would clamp the rotating disks so they couldn't move. The one danger when using a slide rule is that there's a slight movement of the scales while you're turning it to get a look at a needed graduation, leading to a numerical error. This improvement could reduce the chances of that happening.

Another refinement would be a small adjusting screw on each side that could be used to adjust the turning friction of the rotating scales. It appears that a wavy washer is used to provide the friction, but this is non-adjustable. This of course would add to the cost, but that would get amortized out easily over a long period of time. One reason I'd like this feature is that one side of my rule turns a bit too freely and the other side is just perfect -- and everyone has different tastes.

The rotating disks are easy to turn with your thumb while the rule is held in one hand. This leaves your other hand free for e.g. writing with a pencil. You can easily read off squares, square roots, trig values, etc. It's also not difficult to multiply and divide with one hand, although it's easier with two hands.

For older eyes, I'd like the rule scaled up by roughly 20-25% and the font size of the numbers to be increased. This would make it easier to use without a magnifier.

Dislikes

On the trigonometric scales, the small angles and fractions of larger angles are graduated in minutes instead of decimal degrees. For example, in Figure 1, you can see that the graduations between 7° and 8° are in 5 minute intervals. I never met anyone during my career who liked working with sexagesimal measure. It's easy to forget that the scales are marked this way because I just assume they're decimal like all the other scales, then make a mistake when making a calculation. For me, **it's a serious design flaw**.

A minor nit is that the ρ'', ρ', and ρ° marks should be on all the C and D scales, not the one C scale on one side of the rule. The *C* mark is used to get the area of circles from the diameter, so it really only needs to be on the side of the rule with the A and B scales. But since there's an A outer scale and an inner B scale, the *C* mark should be on both the C and D scales. This would let you use what's closest, not have to go looking for it.

References

markup	Markup.pdf at https://someonesdad1.github.io/hobbyutil/project_list.html
post	Frederick Post Company, *The Versalog Slide Rule*, 1951. This is a good instruction manual for using a slide rule. It's aimed primarily at the 10 inch slide rule like the Versalog, but the basic principles will be usable on virtually any slide rule, straight or circular.
srm	https://sliderulemuseum.com/SRM_Home.htm